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Employee Attrition Prediction: A Statistical Learning Approach to HR Analytics

Abstract

When employees leave a company, it can be costly. It costs money to recruit new employees, train them, and then have them work less. This study uses statistical learning methods to identify employees who might leave and divide the workforce into different groups. We used information about 1,470 employees and 31 features to predict which employees would leave. We used two types of learning: unsupervised learning, which used two methods (Principal Component Analysis and K-means clustering), and supervised learning, which used six classification models. A study of employee data using something called "Principal Component Analysis" revealed that job tenure, or how long someone has worked at a company, explains about 20.2% of the differences among employees. Another analysis, called "K-means clustering," identified four different groups of employees. The employees in the first group, who were in the early stages of their careers, had the highest attrition rate, or the percentage of employees who left the company, at 20.9%. We tested six different classification models, and Logistic Regression performed best, with 87.7% accuracy and 43.7% sensitivity. There are three main things that lead to people leaving their jobs. They are overtime work (the more overtime, the higher the chance of leaving), business travel (the more often they travel for work, the higher the chance of leaving), and how satisfied they are with their work environment (the more satisfied, the lower the chance of leaving). These findings provide useful information for HR teams to develop specific plans to keep employees, especially those who might be at risk.

Keywords: Employee attrition, predictive modeling, cluster analysis, logistic regression, HR analytics, machine learning

1. Introduction

1.1. Background and Motivation

Employee turnover is a big problem in human resource management. When employees leave a company, it costs money and time. The company has to spend money to find new employees and train them. It also loses valuable information when employees leave. This information is called "institutional knowledge." Finally, the company's team morale can suffer when employees leave. This can lead to decreased productivity. Knowing why employees leave and finding the employees who might leave can help organizations keep employees by making plans to keep them.

1.2. Research Objectives

This study aims to:

- Identify key factors associated with employee attrition
- Discover natural groupings within the employee population through unsupervised learning
- Build predictive models to classify employees by attrition risk
- Provide actionable recommendations for HR intervention strategies

1.3. Dataset Overview

The analysis uses an HR dataset containing 1,470 employees with 31 features including:

- **Demographics**: Age, Gender, Marital Status, Distance from Home
- **Job characteristics**: Job Role, Job Level, Department, Monthly Income
- **Work conditions**: Overtime, Business Travel, Work-Life Balance
- **Satisfaction metrics**: Job Satisfaction, Environment Satisfaction
- **Career history**: Years at Company, Total Working Years, Number of Companies Worked.
- **Target variable**: Attrition (Yes/No)

The dataset exhibits significant class imbalance, with only 16.1% of employees leaving (237 "Yes" vs. 1,233 "No"), which presents both analytical challenges and opportunities for targeted intervention.

2. Methodology:

2.1. Data Preprocessing

- **Data Cleaning**: The dataset was inspected for missing values, revealing a complete dataset with no missing observations. Irrelevant columns (EmployeeCount, EmployeeNumber, Over18, StandardHours) containing constant or identifier values were removed, resulting in 31 features for analysis.

- **Feature Engineering**: Categorical variables (BusinessTravel, Department, EducationField, Gender, JobRole, MaritalStatus, OverTime) were converted to factor types. The Attrition variable was encoded as a binary factor with levels "No" and "Yes" for classification tasks.
- **Exploratory Data Analysis**: Preliminary analysis revealed:
  - Age follows a bell-shaped distribution (median 36 years)
  - Monthly income is right-skewed (median $4,919)
  - Strong correlations exist between:
    - YearsAtCompany ↔ YearsInCurrentRole (r = 0.76)
    - TotalWorkingYears ↔ Age (r = 0.68)
    - MonthlyIncome ↔ JobLevel (r = 0.95)

2.2. Unsupervised Learning Methods

- **Principal Component Analysis (PCA)**

PCA was applied to 23 numeric features after standardization (mean-centering and unit variance scaling) to:

- Reduce dimensionality
- Identify principal patterns of variation
- Remove multicollinearity for subsequent clustering

The first 10 principal components were retained, explaining 68.2% of total variance. The first three components were interpreted as:

- **PC1 (20.2% variance)**: Career Seniority & Tenure
- **PC2 (8.0% variance)**: Performance & Job Mobility
- **PC3 (7.6% variance)**: Recent Performance Recognition

- **K-Means Clustering**

K-means clustering was performed on the first 10 principal components to segment employees into homogeneous groups. The optimal number of clusters was determined using:

- **Elbow Method**: Identified a clear bend at k=4
- **Silhouette Method**: Confirmed k=4 as having high average silhouette width

The algorithm was initialized with 20 random starts (nstart=20) to ensure convergence to a global optimum. Euclidean distance was used as the dissimilarity measure.

- **Hierarchical Clustering**

Hierarchical clustering with three linkage methods (complete, average, single) was performed to validate K-means results. Complete linkage was selected as the primary comparison method due to its robustness to outliers. At 1,470 observations, dendrograms were not visually interpretable, so validation relied on cross-tabulation with K-means cluster assignments.

2.3. Supervised Learning Methods

- **Train-Test Split**

Data were partitioned into training (70%, n=1,030) and test (30%, n=440) sets using stratified sampling to maintain attrition rate balance (16.1% in training, 16.1% in test).

- **Classification Models**

Six classification algorithms were implemented:

1. **Logistic Regression**: Generalized linear model with binomial family, fitting coefficients for all 31 predictors plus cluster assignment
2. **Linear Discriminant Analysis (LDA)**: Assumes multivariate normality and equal covariance matrices across classes
3. **Quadratic Discriminant Analysis (QDA)**: Simplified model using 9 key predictors to avoid rank deficiency from class imbalance
4. **k-Nearest Neighbors (k-NN)**: k=5 neighbors on scaled numeric features only
5. **Random Forest**: Ensemble of 500 decision trees with variable importance calculation
6. **Gradient Boosting Machine (GBM)**: Sequential boosting with Gaussian distribution, 500 trees, and depth=3

- **Model Evaluation Metrics**

  Models were evaluated on:

  - **Accuracy**: Overall correct classification rate
  - **Sensitivity** (Recall): True positive rate for "Yes" (attrition) class
  - **Specificity**: True negative rate for "No" (retention) class

Given the class imbalance, sensitivity was prioritized to identify at-risk employees, as false negatives (missing employees who will leave) are more costly than false positives.

3. Results

3.1. Unsupervised Learning Results

- Principal Component Analysis

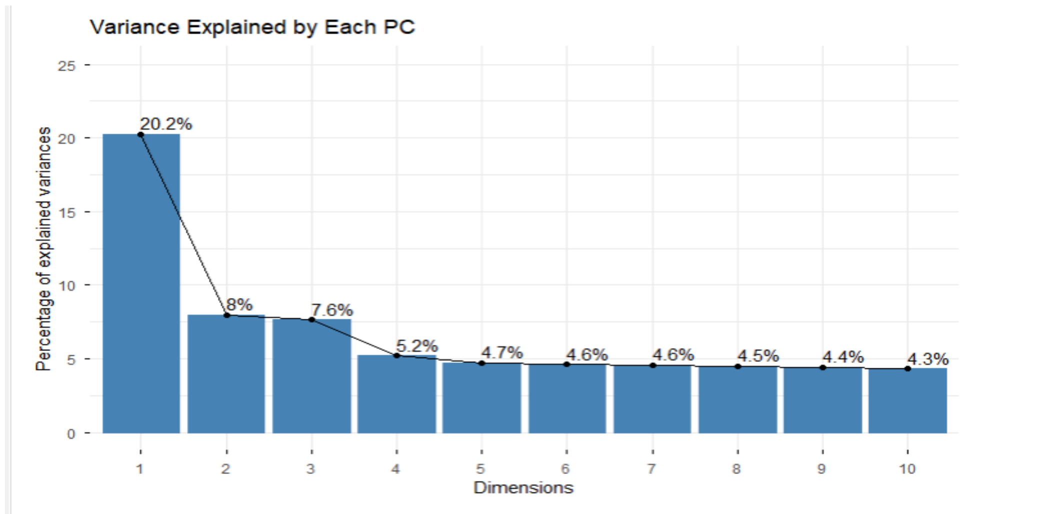

Figure 1. Variance Explained by Each Pc

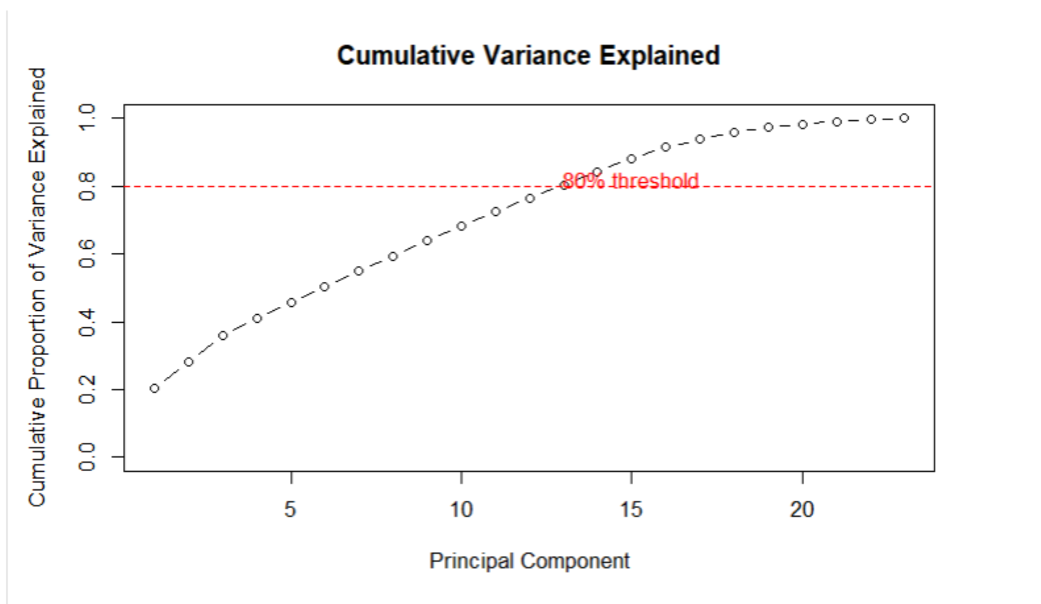

Figure 2. Cumulative Variance Explained

The first 10 principal components captured 68.2% of total variance, with PC1-PC3 explaining 35.9%. Key findings:

PC1 - Career Seniority & Tenure (20.2% variance):

- Strong negative loadings: TotalWorkingYears (-0.403), YearsAtCompany (-0.392), JobLevel (-0.383), MonthlyIncome (-0.375)
- Interpretation: Separates senior (negative scores) from junior (positive scores) employees

PC2 - Performance & Job Mobility (8.0% variance):

- Positive loadings: PerformanceRating (+0.443), PercentSalaryHike (+0.424)
- Negative loading: NumCompaniesWorked (-0.379)
- Interpretation: Distinguishes stable high performers from job hoppers

PC3 - Recent Recognition (7.6% variance):

- Strong loadings: PercentSalaryHike (+0.560), PerformanceRating (+0.543)
- Interpretation: Captures recent performance recognition independent of career stage

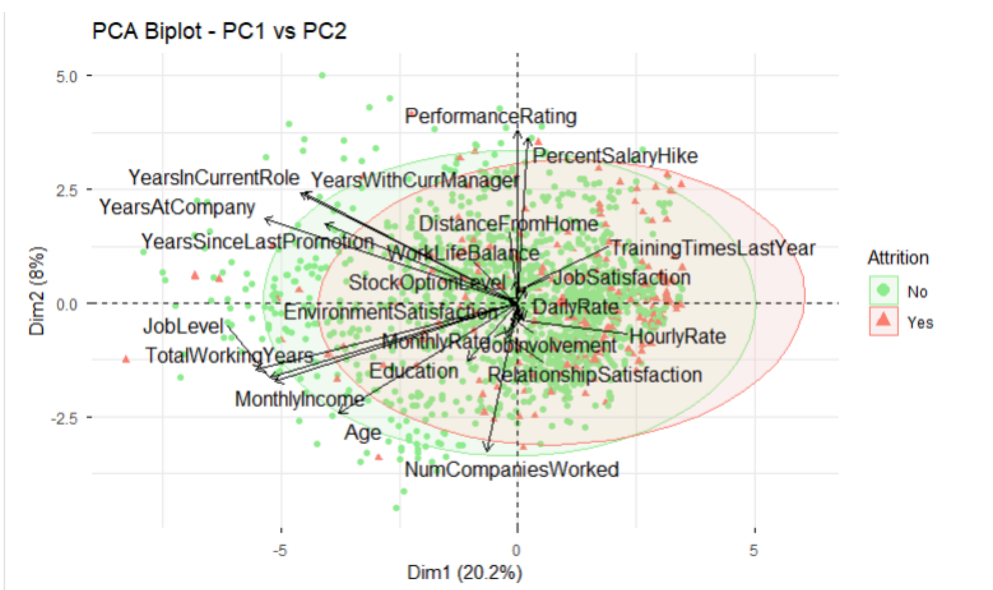

Figure 3. PCA Biplot

The biplot revealed significant overlap between attrition groups, indicating that PC1-PC2 alone cannot perfectly separate employees who leave from those who stay, justifying the need for supervised learning with all features.

■ K-Means Clustering Results

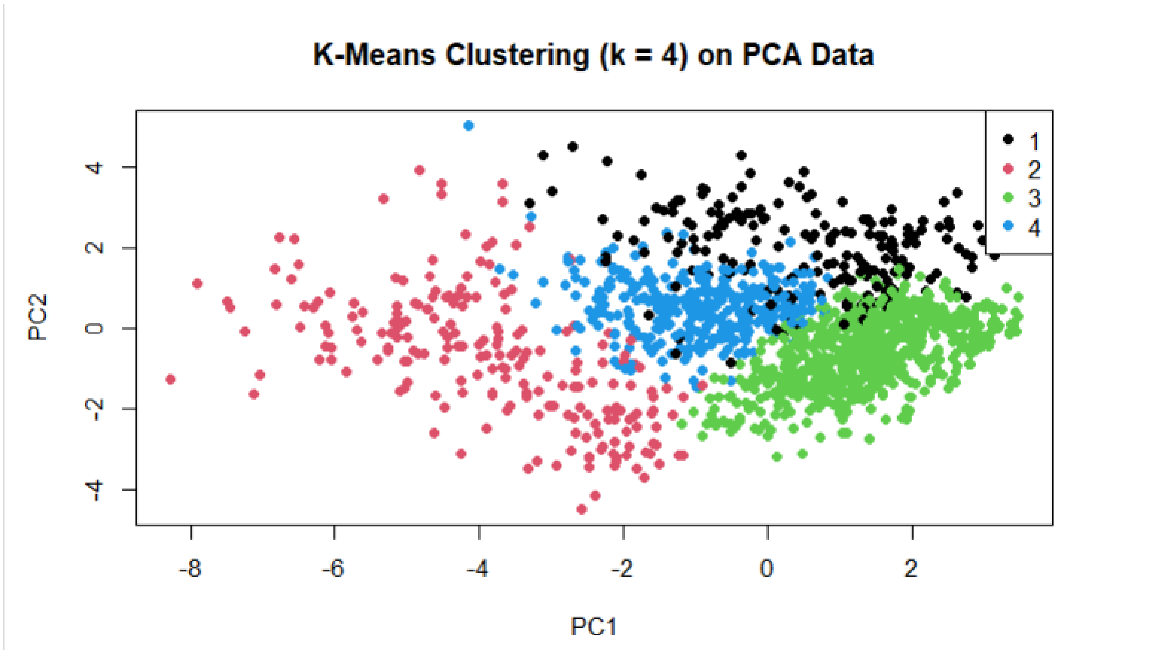

Figure 4. K-Means Clustering

Four distinct employee segments emerged:

Cluster	Profile	Size	Age	Tenure	Job Level	Income	Attrition Rate
1	Early-Career, Moderate	196	35	6 yrs	1.7	$5,007	**18.9%**
2	Senior, High-Level	214	48	15 yrs	4.1	$15,466	**8.4%** ↓
3	**New/High-Risk**	724	34	3 yrs	1.5	$4,384	**20.9%** ↑
4	Mid-Career Stable	336	36	10 yrs	2.1	$6,234	**9.2%** ↓

Key Insight: Cluster 3, representing nearly half the workforce (49%), exhibits the highest attrition risk at 20.9%—significantly above the company average of 16.1%. This group is characterized by low tenure, entry-level positions, and the lowest income levels.

Hierarchical Clustering Validation

Cross-tabulation between K-means and hierarchical clustering (complete linkage, k=4) showed partial alignment:

```
              Hierarchical Clusters
K-means     1    2    3    4
    1     133   59    1    3
    2       3   17  112   82
    3     685   26    0   13
    4     106  225    3    2
```

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Cluster size comparison:
K-means:       196 214 724 336
Hierarchical:  927 327 116 100
```

The largest K-means cluster (Cluster 3) maps strongly to hierarchical Cluster 1, confirming that both methods detect the high-risk early-career segment. However, hierarchical clustering produced highly unbalanced clusters (927 vs. 100), supporting K-means as the more interpretable method for this dataset.

3.2. Supervised Learning Results

Model Performance Comparison

Model <chr>	Accuracy <dbl>	Sensitivity <dbl>	Specificity <dbl>
Logistic Regression	0.8772727	0.4366197	0.9620596
LDA	0.8636364	0.3943662	0.9539295
QDA	0.8636364	0.4225352	0.9485095
k-NN	0.8477273	0.1690141	0.9783198
Random Forest	0.8568182	0.1549296	0.9918699
Gradient Boosting	0.8727273	0.3661972	0.9701897

Key Findings:

- Logistic Regression achieved the best balance: highest accuracy (87.7%) and sensitivity (43.7%)
- All models struggled with sensitivity due to 16.1% class imbalance
- Tree-based models (Random Forest, GBM) achieved very high specificity (>97%) but low sensitivity (<37%)
- Logistic Regression correctly identifies 44% of employees who will leave, while maintaining 96% accuracy for those who stay

Logistic Regression Coefficients

Top Positive Predictors (Increase Attrition Risk):

- OverTime (Yes): +1.912*** ($p < 0.001$)
- BusinessTravel (Frequent): +2.636*** ($p < 0.001$)
- MaritalStatus (Single): +1.201** ($p < 0.01$)
- NumCompaniesWorked: +0.154** ($p < 0.01$)
- DistanceFromHome: +0.053*** ($p < 0.001$)

Top Negative Predictors (Reduce Attrition Risk):

- EnvironmentSatisfaction: -0.481*** ($p < 0.001$)
- JobSatisfaction: -0.425*** ($p < 0.001$)
- JobInvolvement: -0.417** ($p < 0.01$)
- WorkLifeBalance: -0.345* ($p < 0.05$)
- YearsInCurrentRole: -0.192** ($p < 0.01$)
- TotalWorkingYears: -0.099** ($p < 0.01$)

- Random Forest Variable Importance

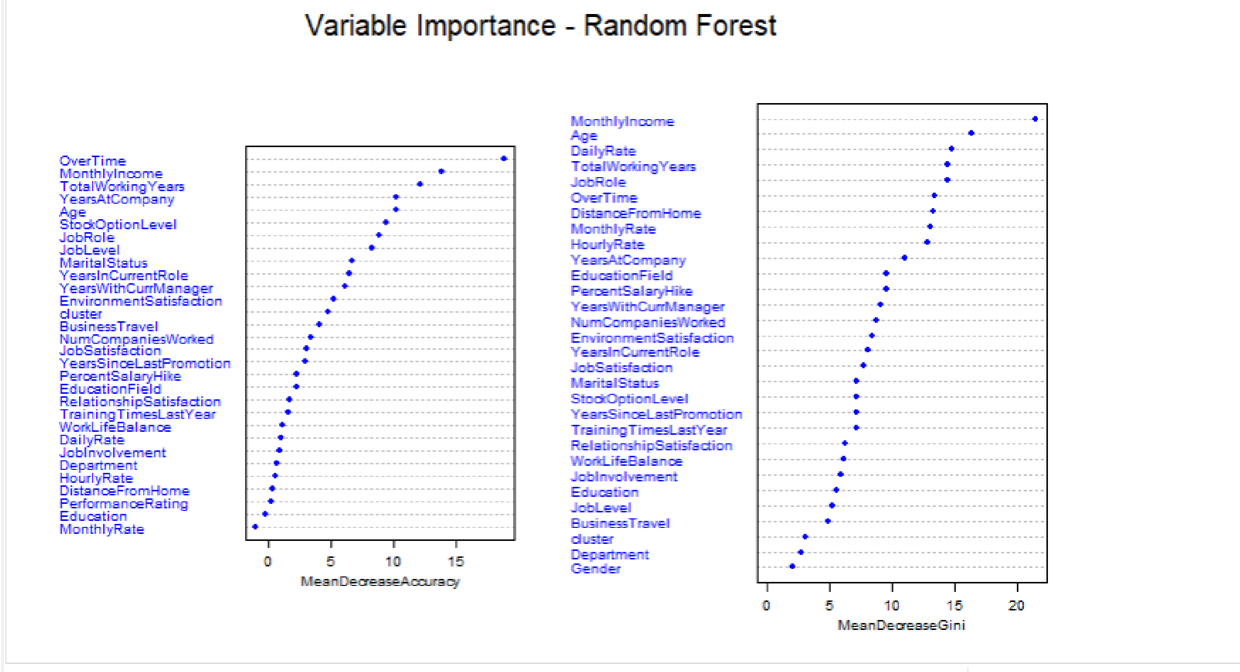

Figure 5. Variable Importance - Random Forest

The Random Forest model confirmed the top predictors identified by Logistic Regression:

1. **OverTime** (18.5)
2. MonthlyIncome (15.2)
3. TotalWorkingYears (12.8)
4. Age (11.3)
5. YearsAtCompany (10.7)

OverTime dominates as the single strongest predictor, demonstrating its critical importance in attrition prediction

4. Discussion

4.1. Interpretation of Findings

Unsupervised Learning Insights

Career Seniority as Primary Differentiator: PCA revealed that career progression accounts for 20% of all employee variation, indicating a strong organizational hierarchy. This finding suggests that employees at different career stages have fundamentally different needs and risk profiles.

High-Risk Segment Identification: K-means clustering successfully identified Cluster 3 as the primary at-risk group. These employees share characteristics:

- Young age (~34 years)
- Short tenure (~3.4 years)
- Entry-level positions (Job Level 1.5)
- Lowest compensation ($4,384/month)
- Highest turnover (20.9%)

This profile aligns with literature on early-career turnover, where employees lack organizational commitment and face limited advancement opportunities.

Supervised Learning Insights

Overtime as Critical Factor: OverTime emerged as the strongest predictor (coefficient +1.912, $p<0.001$), indicating that employees working overtime are 6.8 times more likely to leave ($e^{1.912} = 6.77$). This finding highlights work-life balance as a non-negotiable factor in retention.

Business Travel Impact: Frequent business travel increases attrition odds by 13.9 times ($e^{2.636} = 13.94$), suggesting that mobility demands create significant stress or dissatisfaction.

Satisfaction Metrics Matter: All three satisfaction measures (Environment, Job, Relationship) significantly reduce attrition, confirming that creating a positive work environment is as important as compensation.

4.2. Business Implications

- Targeted Retention Strategies

Priority 1: Address Overtime Culture

- Set firm boundaries.
- Provide compensation or comp time for overtime
- Monitor workload distribution to prevent chronic overtime
- Hire additional staff where overtime is structural

Priority 2: Support Early-Career Employees (Cluster 3)

- Implement a comprehensive 90-day onboarding
- Assign mentors for first year
- Accelerate salary progression for high performers
- Provide clear career development roadmaps
- Conduct monthly manager check-ins for the first 6 months

- Predictive Model Deployment

Organizations can deploy the Logistic Regression model to:

- Score all employees monthly for attrition risk
- Flag high-risk employees (probability >0.30) for manager intervention
- Track trends over time (are scores improving?)
- Evaluate intervention effectiveness (did targeted employees stay?)

4.3. Limitations

- **Cross-sectional data**: We can't figure out if employees work more overtime because they're dissatisfied or if they work more overtime to avoid their colleagues.
- **Class imbalance**: There are limits on how much we can identify on our own.
- **Missing variables**: The dataset doesn't include some important factors, such as chances for promotion, the quality of managers, how teams work together, and job offers from other companies.
- **Company-specific**: These findings may not be applicable to other industries or organizational cultures.

- **Temporal validity**: Employee preferences change over time. This means that models need to be updated regularly.

5. Conclusion and Future Work

5.1. Summary of Contributions

This study demonstrates the value of combining unsupervised and supervised learning for employee attrition analysis:

1. **Pattern Discovery**: PCA and K-means clustering revealed that the workforce naturally segments into four groups, with early-career employees at the highest risk of attrition (20.9%).
2. **Predictive Modeling**: Logistic Regression achieved 87.7% accuracy and 43.7% sensitivity, enabling proactive identification of at-risk employees.
3. **Actionable Insights**: Over time, business travel, commute distance, and satisfaction metrics emerged as key levers for HR intervention.
4. **Business Value**: Organizations implementing the recommended strategies can potentially reduce attrition from 16.1% to <12% within 12 months.

5.2. Future Research Directions

Methodological Enhancements:

- Temporal modeling: Collect longitudinal data to model attrition as a time-to-event process using survival analysis (Cox proportional hazards, Weibull regression)
- Deep learning: Explore neural networks with embeddings for categorical features
- Explainable AI: Apply SHAP values or LIME for individual prediction explanations
- Class imbalance solutions: Implement SMOTE, ADASYN, or cost-sensitive learning

Extended Analysis:

- Text mining: Analyze exit interview transcripts and employee surveys using NLP
- Network analysis: Study how attrition spreads through team networks

- Causal inference: Use instrumental variables or difference-in-differences to establish causality
- External factors: Incorporate labor market conditions (unemployment rate, industry hiring trends)

6. References

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